

Statistics Basics

Data Science / AI Fundamentals

Descriptive Statistics

Before drawing any conclusion from data, you need to describe what's actually in it. Measures of central tendency — mean (average), median (the middle value when sorted, resistant to outliers), and mode (most frequent value) — summarize where data is centered. Measures of spread — range, variance, and standard deviation — describe how much data varies around that center. A dataset with the same mean can look completely different depending on its spread, which is why reporting a mean alone, without any measure of spread, often hides more than it reveals.

Probability Fundamentals

Probability quantifies uncertainty — the likelihood of an event occurring, expressed as a number between 0 (impossible) and 1 (certain). Core concepts that come up constantly in data work: independent events (one event's outcome doesn't affect another's probability), conditional probability (the probability of an event given that another has already happened, written $P(A|B)$), and the difference between correlation and causation — two variables moving together does not, by itself, mean one causes the other. This last distinction is one of the most commonly violated principles in casual data analysis, and catching it is a genuinely valuable skill.

Distributions

A distribution describes how values in a dataset are spread across their possible range. The normal (Gaussian) distribution — the familiar bell curve, symmetric around its mean — appears constantly in nature and is the foundation for many statistical methods that assume data is roughly normally distributed. Other common distributions include the uniform distribution (every value in a range equally likely) and skewed distributions (where data clusters toward one end, common in real-world data like income or wait times). Visualizing a variable's distribution (via a histogram) before analyzing it further is a habit that catches a surprising number of data-quality problems early.

Hypothesis Testing Basics

Hypothesis testing is the formal process of using sample data to decide whether an observed effect is likely real or could plausibly be due to random chance. The core idea: state a null hypothesis (there's no real effect), collect data, and calculate a p-value (roughly, how surprising the observed data would be if the null hypothesis were actually true). A small p-value (commonly below 0.05, though that threshold is a convention, not a law of nature) suggests the

observed effect is unlikely to be pure chance. This is genuinely subtle — a low p-value doesn't prove an effect is real or important, only that it's unlikely under the null hypothesis, and it says nothing about practical significance.

Learn More

Khan Academy — Statistics and Probability

<https://www.khanacademy.org/math/statistics-probability>

NPTEL — Statistics for Data Science

<https://nptel.ac.in/>

StatQuest YouTube Channel

<https://www.youtube.com/@statquest>